Birzeit University
Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
ENCS3320 – Computer Networks (Term 1251)
Project #2 (*Cisco Packet Tracer*) – Due 18/1/2026

A) Objectives

This project aims to enhance your knowledge and practical skills in computer network design, including subnetting and routing. The specific objectives are:

- Design an IP addressing scheme using subnetting.
- Learn how to use Cisco Packet Tracer to design and simulate network topologies.
- Develop skills in configuring routing protocols.
- Learn the fundamentals of wireless Local Area Network (LAN) configuration.
- Gain proficiency in using fundamental network commands.
- Develop professional technical report writing skills to effectively document and communicate project results and analyses.
- Strengthen teamwork skills through collaboration on project tasks.

B) Requirements and Deliverables

As a team of ***three students*** *(from any section)*, your task is to design a network of networks using Cisco Packet Tracer and submit both your **.pkt** project file (named by ***Project2.pkt***) and a project report (***Project2_report.pdf***). Please adhere to the following guidelines:

- **Cisco Packet Tracer Installation:** Download the latest version of Cisco Packet Tracer from https://www.netacad.com/cisco-packet-tracer.
- **Implementation Guidelines:** Ensure that the implementation reflects *your work*. Refer to *Section D (Helpful Tutorials)* for resources to assist with creating the topology.
- **Team Coordination:** Active participation from each team member is essential at every stage of the project, including research, design, implementation, testing, and report preparation.
- **Submission Requirements:** Submit the following **files** in a *single compressed folder* named ***Project2.zip*** through ITC (https://itc.birzeit.edu/course/view.php?id=38380):
  1) **Project Report (*Project2_report.pdf*)**
     A comprehensive technical report in **PDF** format that documents your solutions. The report should include:
     a. **Cover Page**

- Include the university logo, department, course name and number, project title, team members' names and IDs (with their sections), and submission date.

b. **Theory and Procedure**
- Explain the theoretical concepts relevant to each project requirement, organized under appropriate section titles.
- List the components or methods used for each part of the project, with a brief justification for each.
- Use diagrams or flowcharts to illustrate your solutions. Provide detailed descriptions and captions for all figures, ensuring they are referenced in the text.
- Cite all theoretical concepts and sources and include a references section at the end of the report.

c. **Results and Discussions**
- Provide a detailed discussion of your approach and the results for each task outlined in *Section C (Topology & Tasks)*.
- Include clear screenshots that demonstrate the functionality of all required tasks. Each screenshot must display the system's **date** and **time**. Accompany each screenshot with caption, and an explanation, detailing the results and their significance.

d. **Issues and Limitations**
- Highlight challenges, limitations, or issues encountered during the project, whether individual or team-related.
- Note any tasks or components that did not function as expected, if applicable.

e. **Teamwork**
- Use a chart to document each team member's contributions, detailing specific tasks completed by each individual.

Ensure the report includes proper numbering, a table of contents, a list of figures, a list of tables, and appendices as necessary.

2) **Network Topology (*Project2.pkt*)**

Create the network topology shown in ***Figure 1***, adhering to the following requirements:

a. Use the following **devices** in your implementation:
   - Routers (**Router-PT**)
   - Switches (**Switch-PT**)
   - Access Point (**AccessPoint-PT**)
   - PCs/Laptops (**PC-PT/Laptop-PT**)
   - Smartphone (**SMARTPHONE-PT**)

b. Use **automatic connections** to link the devices.

c. For laptops, replace the laptop's wired LAN module with a **WPC300N** wireless adapter module to provide wireless connectivity with the access point.

C) Topology & Tasks:

The network topology comprises ***three*** areas: (i) Central campus network (***Area 0***), (ii) West network (***Area 20***), (iii) East network (***Area 30***). These networks are interconnected through the central network, as depicted in ***Figure 1***. For the IP setting for each network, you are going to do the subnetting task. The main IP (1***X***.***Y***.128.0/17), replace the variable ***X*** and ***Y*** with the last four digits of the team leader's student ID (SID). For example, if the team leader's SID is 123**7090**, then ***X*** = **70 and** ***Y*** = **90**, the IP will be in this example **170.90.128.0/17**. Solutions that *fail to incorporate the SID* into the

network IP addresses as specified will be deemed invalid and **rejected**. The project consists of the following major tasks:

1) **IP Subnetting Task:**

Create the necessary IP addressing and subnet plan using the specified address, the number of devices in **Table 1**, and the topology illustrated in **Figure 1**.

Table 1: Number of hosts for each network.

Network	Subnetwork	Name	Hosts
Central	**LAN**	C-LAN	500
West	**LAN1**	W-LAN1	200
	LAN2	W-LAN2	260
	WLAN	W-WLAN	480
East	**LAN1**	E-LAN1	112
	LAN2	E-LAN2	150
	WLAN	E-WLAN	550

Using the main IP address and the number of hosts in each network, compute the IP required for each subnetwork. A table **should** be included in your report showing the network IP, broadcast IP, range of usable IPs, and the corresponding CIDR notation (/X) for the subnet mask.

2) **Building Topology Task:**

The structure of the central network is shown in **Figure 1** and subnet IP addresses for this network should be assigned based on the results of **IP Subnetting Task.** Here the details of the networks:

a) Central Campus network (Area 0):

It consists of one subnetwork (**Central LAN**), follow these steps to build and configure the network:

1. Add two routers (Router-PT).
2. Add one switch (Switch-PT)
3. Add two end devices: 1 PC (PC-PT) and 1 laptop (Laptop-PT). Assign static IP addresses as follows: PC is given the IP address of the 200th host in the subnet. Laptop is the 350th.
4. Turn on the interfaces and assign IPs.
5. Do the requested routing as in the **Routing Configuration Part**.

b) West network (Area 20):

It consists of three subnetworks (**West LAN1, West LAN2** and **West WLAN)**, follow these steps to build and configure the network:

1. Add three routers (Router-PT)
2. Each interface has an IP address based on the subnetting part.
3. Turn on the interfaces and assign IPs.
4. Add two switches (Switch-PT).
5. Add one wireless access point (AccessPoint-PT).
6. For subnetwork (**West LAN1**), add two end devices: 1 PC (PC-PT) and 1 laptop (Laptop-PT). Assign static IP addresses as follows: PC is given the IP address of the 50th host in the subnet. Laptop is the 180th.
7. For subnetwork (**West LAN2**), on the second switch, add two end devices: 1 PC (PC-PT) and 1 laptop (Laptop-PT). Assign static IP addresses as following: PC is given the IP address of the 114th host in the subnet. Laptop is the 258th.

8. For subnetwork (**West WLAN**), add the wireless access point. To this wireless network, add one laptop (Laptop-PT), and one smartphone (Smartphone-PT). Here are the settings of the access point:
   - **Name**: ENCS3320WestSTName
   - **Security Model**: WPA2 Personal
   - **Encryption Type**: AES
   - **Key**: ST_# (e.g., ST_1237090)

   Assign static IP addresses as follows: PC is given the IP address of the 80th host in the subnet. Laptop is the 438th.

c) East network (Area 30):

It consists of three subnetworks (**East LAN1, East LAN2** and **East WLAN)**, follow these steps to build and configure the network:

1. Add three routers (Router-PT)
2. Each interface has an IP address based on the subnetting part.
3. Turn on the interfaces and assign IPs.
4. Add two switches (Switch-PT).
5. Add one wireless access point (AccessPoint-PT).
6. For subnetwork (**East LAN1**), on the first switch, add two end devices: 1 PC (PC-PT) and 1 laptop (Laptop-PT). Assign static IP addresses as following: PC is given the IP address of the 30th host in the subnet. Laptop is the 110th.
7. For subnetwork (**East LAN2**), on the second switch, add two end devices: 1 PC (PC-PT) and 1 laptop (Laptop-PT). Assign static IP addresses as following: PC is given the IP address of the 70th host in the subnet. Laptop is the 145th.
8. For subnetwork (**East WLAN**), add the wireless access point. To this wireless network, add one laptop (Laptop-PT), and one smartphone (Smartphone-PT). Here are the settings of the access point:
   - **Name**: ENCS3320EastSTName
   - **Security Model**: WPA2 Personal
   - **Encryption Type**: AES
   - **Key**: ST_# (e.g., ST_1237090)

   Assign static IP addresses as follows: PC is given the IP address of the 15th host in the subnet. Laptop is the 540th.

d) ISP network:

It consists of one subnetwork, follow these steps to build and configure the network:

1. Turn on the interface and assign IP.
2. Add one router (Router-PT) to be the ISP router.

3) **Routing Configuration Task:**

- Use open shortest path first (**OSPF**) for *routing* on the three given areas in the topology (**Area 0**, **Area 20** and **Area 30**). You need to do the following on all routers:
  a. Configuring OSPF on a router:
     - **`Router(config)# router ospf <Process-ID>`**
       This command starts the OSPF routing process with the process number Process-ID. It is recommended to use value 1 for the process number on all routers.
  b. Adding networks to the OSPF protocol:

- **`Router(config-router# network <ID-Address> <WildCard-Mask> area <Area-ID>`**
  This command defines an interface on which OSPF runs and defines the area ID for that interface.

- Use OSPF routing for ISP network also.

4) **Testing and Troubleshooting Task:**

You need to do the following and show them in your report:

a) Include labels on the topology in the **.pkt** file for clarity.

b) Include the following screenshots in your project report:
   - Static IP configuration for *routers*.
   - IP configuration for the assigned end devices.
   - Successful **ping** and **tracert** results between end devices.
   - Routing configuration using show **ip route** command.

D) Helpful Tutorials

- Cisco Packet Tracer download and installation: https://www.youtube.com/watch?v=5dqZ623N74
- Access Point configuration: https://www.youtube.com/watch?v=-E4yivI26SY
- OSPF routing configuration: https://www.youtube.com/watch?v=1iJ882Xqqm0

E) Grading Criteria

The total weight is 7% of the course grade. Performance on this project will be evaluated competitively based on achieving the required tasks.

Each team must submit **one** final version before the deadline. **Late submissions *will not be accepted*** under any circumstances. Failure to submit your project solutions before the deadline will result in a score of zero.

GOOD LUCK

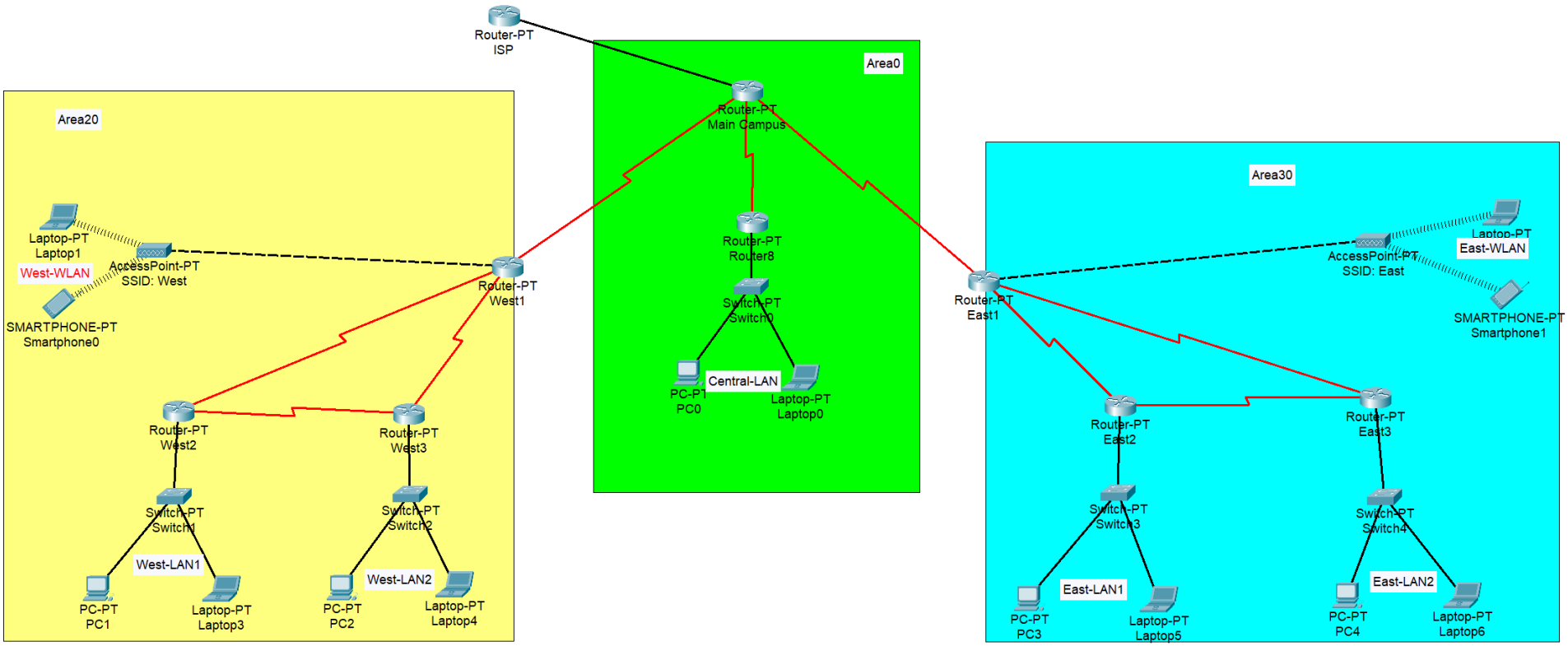

Figure 1: Network Topology.